

In an SDE placement MCQ round, the goal is often to filter for candidates who understand the core terminology and the "why" behind system behavior.⁴ This updated list covers the five primary pillars of Operating Systems at a foundational level.

Section I: OS Basics and System Protection

1. What is the primary purpose of an Operating System?

- A. To design web pages and graphics.
- B. To act as an intermediary between the user and the computer hardware.
- C. To compile high-level code into assembly.
- D. To increase the physical storage capacity of the hard drive.
- **Answer: B**
- **Explanation:** The OS manages hardware resources (CPU, Memory, I/O) and provides an environment for application software to run.

2. Which of the following is considered the "heart" or core component of the OS?

- A. The Shell.
- B. The Compiler.
- C. The Kernel.
- D. The Command Prompt.
- **Answer: C**
- **Explanation:** The kernel is the most essential part of the OS, managing system resources and communication between hardware and software.

3. What is a "System Call"?

- A. A call to a remote server via the internet.
- B. A mechanism for programs to request services from the OS kernel.
- C. An error message displayed when a program crashes.
- D. A hardware signal to increase CPU speed.
- **Answer: B**
- **Explanation:** Applications run in restricted user mode and must use system calls to perform privileged tasks like reading a file or allocating memory.

4. What is the main difference between "User Mode" and "Kernel Mode"?

- A. Kernel mode is slower than user mode.
- B. User mode has full access to hardware; kernel mode does not.
- C. Kernel mode has unrestricted access to hardware; user mode is restricted.
- D. There is no technical difference.
- **Answer: C**
- **Explanation:** Dual-mode operation protects the system by preventing user applications from accidentally or intentionally crashing the hardware.

5. Which type of OS is designed to guarantee a response within a fixed, strict time limit?

- A. Batch OS.
- B. Real-Time OS (RTOS).
- C. Distributed OS.
- D. Time-Sharing OS.
- **Answer: B**
- **Explanation:** RTOS is used in time-critical environments like medical devices, flight controllers, or industrial robotics.⁶

6. What is "Multiprogramming"?

- A. Running a single program on multiple CPUs.
- B. Keeping multiple programs in main memory so the CPU always has something to execute.
- C. Writing programs in multiple languages.
- D. Using multiple users on a single computer.
- **Answer: B**
- **Explanation:** Multiprogramming increases CPU utilization by ensuring the processor never stays idle while one process is waiting for I/O.⁹

7. Which component acts as a "command interpreter" for the OS?

- A. Kernel.
- B. Shell.
- C. BIOS.
- D. Registry.
- **Answer: B**
- **Explanation:** The Shell provides the interface (like Bash or CMD) for the user to enter commands that the OS then executes.

8. What is "Bootting" in Operating Systems?

- A. Deleting all files from the hard drive.
- B. The process of starting or restarting the computer and loading the OS.
- C. Increasing the RAM of the system.
- D. Running a program for the first time.
- **Answer: B**
- **Explanation:** Booting (or bootstrapping) involves the hardware loading the OS kernel into memory from the disk.

9. Which of these is NOT a function of the Operating System?

- A. Memory Management.
- B. File Management.

- C. Compiling Source Code.
- D. Device Management.
- **Answer: C**
- **Explanation:** Compiling is a task for a Compiler, which is application or system software, but not a core part of the OS itself.

10. What is "Time-Sharing"?

- A. Dividing a task among multiple users.
- B. A technique that allows many users to share computer resources simultaneously via rapid CPU switching.
- C. Allowing two computers to share the same RAM.
- D. Using the computer only during specific hours.
- **Answer: B**
- **Explanation:** Time-sharing is a logical extension of multiprogramming that allows interactive user sessions.⁹

Section II: Process and Thread Management

11. What is a "Process"?

- A. A static file on the hard drive.
- B. A program in execution.
- C. A type of hardware device.
- D. A set of rules for networking.
- **Answer: B**
- **Explanation:** While a program is passive (on disk), a process is an active entity currently being managed by the CPU and RAM.

12. What is a "Thread"?

- A. A separate process running on a different computer.
- B. A "lightweight process" that shares resources with other threads in the same process.
- C. A physical wire inside the CPU.
- D. A method for saving files.
- **Answer: B**
- **Explanation:** Threads are the smallest unit of execution and are faster to create and switch than processes because they share the same memory space.

13. In which state is a process when it is waiting for the CPU to be assigned?

- A. Running.
- B. Waiting.
- C. Ready.
- D. New.

- **Answer: C**
- **Explanation:** A process in the "Ready" state has all the resources it needs and is simply waiting for its turn on the processor.

14. What does the "Process Control Block" (PCB) store?

- A. The user's personal files.
- B. Information required by the OS to manage a specific process (ID, state, registers).
- C. The source code of the operating system.
- D. A list of all websites visited.
- **Answer: B**
- **Explanation:** Every process has a PCB that acts as its "ID card," storing everything the OS needs to pause and resume that process.

15. What is a "Context Switch"?

- A. Changing the user account.
- B. Switching the system off.
- C. Saving the state of a current process and loading the state of a new one.
- D. Moving a file from one folder to another.
- **Answer: C**
- **Explanation:** Context switching is the overhead task that allows an OS to switch between multiple tasks seamlessly.

16. Which CPU scheduling algorithm follows the "First Come, First Served" logic?

- A. Round Robin.
- B. SJF.
- C. FCFS.
- D. Priority.
- **Answer: C**
- **Explanation:** FCFS is the simplest algorithm where processes are executed in the exact order they arrive.

17. What is the main characteristic of "Round Robin" scheduling?

- A. It always picks the shortest task.
- B. It assigns a fixed "time quantum" (time slice) to each process.
- C. it executes tasks in reverse order.
- D. It only works on multi-core CPUs.
- **Answer: B**
- **Explanation:** Round Robin is designed for time-sharing systems to ensure fairness and responsiveness.

18. What is "Starvation" in the context of priority scheduling?

- A. A process crashing due to memory leak.
- B. A low-priority process never getting the CPU because high-priority tasks keep arriving.
- C. The system shutting down due to low power.
- D. The CPU running out of instructions.
- **Answer: B**
- **Explanation:** Starvation is a common problem in priority systems where "weak" tasks are perpetually ignored.

19. "Aging" is a solution used to prevent:

- A. Thrashing.
- B. Starvation.
- C. Deadlocks.
- D. Fragmentation.
- **Answer: B**
- **Explanation:** Aging gradually increases the priority of processes that have been waiting in the queue for a long time.

20. In the "Waiting" (or Blocked) state, a process is typically waiting for:

- A. The CPU.
- B. An I/O event or signal completion.
- C. The computer to restart.
- D. More RAM to be installed.
- **Answer: B**
- **Explanation:** A process moves to the waiting state when it cannot proceed without external input or output completion.

21. Which scheduling algorithm minimizes the "Average Waiting Time"?

- A. FCFS.
- B. Round Robin.
- C. Shortest Job First (SJF).
- D. Priority.
- **Answer: C**
- **Explanation:** SJF is theoretically optimal for reducing wait time, though it is hard to implement because we don't know the future duration of tasks.

22. Preemptive scheduling means:

- A. A process cannot be interrupted once it starts.
- B. The OS can interrupt a running process to give the CPU to another.
- C. The process determines its own priority.
- D. Only one process can ever run on the system.
- **Answer: B**
- **Explanation:** Preemption allows the OS to maintain control and enforce time-sharing or

handle high-priority interrupts.

23. Which of the following is NOT shared between threads of the same process?

- A. Code.
- B. Data.
- C. Stack and Registers.
- D. Files.
- **Answer: C**
- **Explanation:** Each thread must have its own stack to track its own function calls and its own register values for its current state.

24. What is a "Race Condition"?

- A. Two CPUs competing to see which is faster.
- B. A situation where multiple threads access shared data and the result depends on the order of execution.
- C. A networking error.
- D. A process running too fast for the RAM.
- **Answer: B**
- **Explanation:** Race conditions lead to unpredictable bugs and data corruption if proper synchronization is not used.

25. What is the "Critical Section"?

- A. A segment of the OS that cannot be deleted.
- B. A part of a program where shared resources are accessed.
- C. The startup code of the BIOS.
- D. A protected part of the hard drive.
- **Answer: B**
- **Explanation:** Only one process should be allowed to execute in its critical section at a time to maintain data integrity.

Section III: Deadlock and Synchronization

26. What is a "Deadlock"?

- A. A computer that won't turn on.
- B. A state where two or more processes are waiting forever for each other to release resources.
- C. A process that finished its task successfully.
- D. A disconnected network cable.
- **Answer: B**
- **Explanation:** Deadlock creates a total standstill in the system where no progress is possible.

27. Which of the following is NOT a necessary condition for a deadlock?

- A. Mutual Exclusion.
- B. Circular Wait.
- C. Preemption.
- D. Hold and Wait.
- **Answer: C**
- **Explanation:** Deadlock requires "No Preemption." If the OS could take resources back, deadlocks would be easily broken.

28. What does "Mutual Exclusion" mean?

- A. Multiple processes can share a resource at the same time.
- B. Only one process can use a resource at any given time.
- C. All processes must use the same priority.
- D. The CPU can run two processes at once.
- **Answer: B**
- **Explanation:** This is a core requirement for protecting shared data, but also a core cause of potential deadlocks.

29. A "Semaphore" is:

- A. A hardware device for networking.
- B. A synchronization tool (integer variable) used to control access to shared resources.
- C. A type of CPU cache.
- D. A file compression method.
- **Answer: B**
- **Explanation:** Semaphores help solve synchronization problems by signaling when resources are busy or free.

30. The "Banker's Algorithm" is used for:

- A. Deadlock Detection.
- B. Deadlock Prevention.
- C. Deadlock Avoidance.
- D. CPU Scheduling.
- **Answer: C**
- **Explanation:** It "avoids" deadlocks by checking if granting a resource request keeps the system in a "safe state."

31. What is a "Safe State"?

- A. When the computer is in "Safe Mode."
- B. When the OS can allocate resources to all processes in some order without a deadlock occurring.
- C. When no processes are running.

- D. When the firewall is turned on.
- **Answer: B**
- **Explanation:** A safe state guarantees that a deadlock can be avoided even if all processes request their maximum resources.¹¹

32. The "Circular Wait" condition can be prevented by:

- A. Increasing the RAM.
- B. Defining a linear order of resource types.
- C. Using only one process.
- D. Turning off the computer.
- **Answer: B**
- **Explanation:** If resources must be requested in a specific order (e.g., Disk then Printer), a cycle of waiting can never form.

33. What is a "Mutex"?

- A. A multi-tasking OS.
- B. A binary semaphore used for mutual exclusion.
- C. A type of memory management.
- D. A hardware error.
- **Answer: B**
- **Explanation:** A Mutex (Mutual Exclusion) allows only one thread to access a resource at a time, acting like a lock.

34. "Busy Waiting" refers to:

- A. The user waiting for a website to load.
- B. A process looping repeatedly to check if a lock is available.
- C. The CPU performing background updates.
- D. A fast way to save files.
- **Answer: B**
- **Explanation:** Busy waiting is inefficient because the process consumes CPU cycles just to "wait" in a loop.

35. A deadlock can be "detected" using which tool?

- A. Gantt Chart.
- B. Resource Allocation Graph.
- C. Flowchart.
- D. Binary Search.
- **Answer: B**
- **Explanation:** A cycle in a Resource Allocation Graph (for single-instance resources) indicates a deadlock.

Section IV: Memory Management (Based on Sanfoundry Fundamentals)

36. "Logical Address" is an address generated by:

- A. The RAM.
- B. The CPU.
- C. The Hard Drive.
- D. The BIOS.
- **Answer: B**
- **Explanation:** Programs work with logical (virtual) addresses, which the OS then maps to real physical addresses in hardware.

37. What is "Paging"?

- A. A method of organizing data on a disk.
- B. A memory management scheme that eliminates the need for contiguous allocation of physical memory.
- C. Increasing the page file size on the disk.
- D. Turning pages in a digital document.
- **Answer: B**
- **Explanation:** Paging allows a process's memory to be scattered across non-contiguous "frames" in RAM, solving external fragmentation.

38. In paging, physical memory is broken into fixed-sized blocks called:

- A. Pages.
- B. Segments.
- C. Frames.
- D. Inodes.
- **Answer: C**
- **Explanation:** Frames are the physical units in RAM; Pages are the logical units in the process.

39. When does a "Page Fault" occur?

- A. When the hardware breaks.
- B. When a process tries to access a page that is not currently in physical RAM.
- C. When the system runs out of disk space.
- D. When a file is renamed.
- **Answer: B**
- **Explanation:** A page fault triggers the OS to bring the missing page from the disk's swap space into RAM.

40. What is "External Fragmentation"?

- A. Unused space within an allocated block.
- B. Unused space between blocks of allocated memory.
- C. A virus from an external drive.
- D. When the hard drive is too full.
- **Answer: B**
- **Explanation:** This happens when total free memory is enough for a request, but it is not continuous, so it cannot be used.

41. "Internal Fragmentation" is a drawback of which system?

- A. Segmentation.
- B. Paging.
- C. Virtual Memory.
- D. All of the above.
- **Answer: B**
- **Explanation:** Because paging uses fixed-size blocks, a process might not perfectly fill its last page, leaving some wasted space inside that page.

42. What is "Virtual Memory"?

- A. A type of extremely fast RAM.
- B. A technique that allows the execution of processes larger than physical memory.
- C. Memory used by virtual reality headsets.
- D. A simulation of a computer.
- **Answer: B**
- **Explanation:** Virtual memory uses the hard drive as an extension of RAM to run large or numerous applications.

43. Which component is responsible for translating logical addresses to physical addresses?

- A. ALU.
- B. Memory Management Unit (MMU).
- C. Control Unit.
- D. Scheduler.
- **Answer: B**
- **Explanation:** The MMU is a hardware device that performs the fast translation required at runtime.

44. What is "Thrashing"?

- A. A process that is running very fast.
- B. A situation where the system spends more time swapping pages than executing instructions.

- C. Deleting files too quickly.
- D. A hardware failure in the disk head.
- **Answer: B**
- **Explanation:** Thrashing happens when the total "working set" of all processes is too large for the physical RAM, causing a performance collapse.

45. Which page replacement algorithm replaces the page that arrived first?

- A. LRU.
- B. Optimal.
- C. FIFO (First-In, First-Out).
- D. LIFO.
- **Answer: C**
- **Explanation:** FIFO is simple but can lead to poor performance (and Belady's Anomaly).

Section V: File Systems and I/O Basics

46. What is an "Inode" in UNIX-like systems?

- A. A type of network port.
- B. A data structure that stores file metadata (size, owner, pointers).
- C. The actual content of the file.
- D. A shortcut on the desktop.
- **Answer: B**
- **Explanation:** An inode contains everything about a file except for its name and the actual data.

47. Which Disk Scheduling algorithm serves requests in the order they arrive?

- A. SCAN.
- B. SSTF.
- C. FCFS.
- D. LOOK.
- **Answer: C**
- **Explanation:** FCFS is the simplest disk scheduling policy but involves the most mechanical head movement.

48. What is "Seek Time" for a hard disk?

- A. The time to read the data from a sector.
- B. The time it takes for the disk arm to move to the correct cylinder.
- C. The time for the disk to rotate.
- D. The time the OS takes to find the file name.
- **Answer: B**
- **Explanation:** Seek time is the most expensive mechanical delay in a traditional hard

drive.

49. What is "Spooling"?

- A. A method of memory allocation.
- B. Storing data temporarily in a buffer so a shared device (like a printer) can handle it later.
- C. A way to speed up the internet.
- D. Encrypting a file for security.
- **Answer: B**
- **Explanation:** Spooling allows multiple users to send jobs to a device simultaneously without causing conflicts.

50. "Direct Memory Access" (DMA) is used to:

- A. Increase the CPU clock speed.
- B. Allow I/O devices to transfer data to RAM without using the CPU for every byte.
- C. Access the internet directly.
- D. Prevent users from deleting files.
- **Answer: B**
- **Explanation:** DMA improves performance by letting high-speed devices move large blocks of data while the CPU does other work.

Part I: Introduction & OS Architecture (Questions 1-30)

1. What is the main objective of a multiprogramming operating system?

- A. To provide a better user interface.
- B. To maximize CPU utilization.
- C. To allow multiple users to share a single CPU.
- D. To decrease the hardware cost.
- **Answer: B | Explanation:** Multiprogramming ensures the CPU is never idle by keeping multiple jobs in memory.

2. Which of the following is an example of an "Open Source" OS?

- A. Windows 11.
- B. macOS.
- C. Linux.
- D. iOS.
- **Answer: C | Explanation:** Linux source code is freely available for anyone to view and modify.

3. What is the role of the "Command Interpreter" or Shell?

- A. To manage the physical memory.
- B. To provide the interface between user and kernel by interpreting user commands.
- C. To compile high-level code.
- D. To protect the system from viruses.
- **Answer: B | Explanation:** The shell (like Bash or CMD) takes user inputs and converts them into system-level requests.

4. A Microkernel differs from a Monolithic kernel because it:

- A. Is much larger in size.
- B. Moves non-essential services (like drivers) from kernel space to user space.
- C. Does not support multitasking.
- D. Is only used in mobile devices.
- **Answer: B | Explanation:** Microkernels aim for stability by keeping only core tasks in the kernel.

5. What is a "Trap"?

- A. A hardware failure.
- B. A software-generated interrupt.
- C. A virus that locks the system.
- D. A method for process scheduling.
- **Answer: B | Explanation:** A trap is a software-initiated signal, often caused by an error (like division by zero) or a system call.

6. Which of the following is NOT a type of Operating System?

- A. Real-Time OS.
- B. Batch OS.
- C. Java OS.
- D. Distributed OS.
- **Answer: C | Explanation:** Java is a programming language/platform, not a fundamental category of OS.

7. "Spooling" is most commonly used for which of the following?

- A. Increasing RAM speed.
- B. Handling data for slow devices like printers.
- C. Encrypting system files.
- D. Managing network passwords.

- **Answer: B | Explanation:** Spooling (Simultaneous Peripheral Operations On-Line) allows multiple jobs to be queued for a single peripheral.

8. Which mode of execution prevents user programs from accessing hardware directly?

- A. Supervisor Mode.
- B. Kernel Mode.
- C. User Mode.
- D. Debug Mode.
- **Answer: C | Explanation:** User mode is a restricted state that ensures system security and stability.

9. The process of loading the OS into memory is called:

- A. Swapping.
- B. Bootstrapping (Booting).
- C. Paging.
- D. Spooling.
- **Answer: B | Explanation:** Bootstrapping involves the BIOS/UEFI loading the initial part of the OS into RAM.

10. What is an "API" (Application Programming Interface)?

- A. A hardware chip on the motherboard.
- B. A set of routines and tools for building software applications.
- C. A type of process scheduler.
- D. A method for disk formatting.
- **Answer: B | Explanation:** APIs define how different software components should interact.

11. Which system call is used to create a new process in Unix-like systems?

- A. `exec()`
- B. `fork()`
- C. `wait()`
- D. `exit()`
- **Answer: B | Explanation:** `fork()` creates an exact duplicate (child) of the current parent process.

12. "Time-Sharing" is a logical extension of:

- A. Single-user systems.
- B. Real-time systems.
- C. Multiprogramming.
- D. Batch processing.
- **Answer: C | Explanation:** Time-sharing allows multiple users to interact with a system at once by switching the CPU rapidly between them.

13. In which mode does the OS kernel run?

- A. User Mode.
- B. Privileged/Kernel Mode.
- C. Application Mode.
- D. Virtual Mode.
- **Answer: B | Explanation:** Kernel mode has full access to all hardware and instructions.

14. Which component is responsible for managing the hardware and software communication?

- A. Shell.
- B. Kernel.
- C. Compiler.
- D. Utility program.
- **Answer: B | Explanation:** The kernel is the core bridge between hardware and software.

15. What is "Throughput" in OS?

- A. The time to finish a single task.
- B. The number of tasks completed per unit of time.
- C. The size of the process in RAM.
- D. The speed of the network connection.
- **Answer: B | Explanation:** High throughput is a goal of efficient batch and multiprogramming systems.

16. Which OS is best for systems with strict timing constraints?

- A. Batch OS.
- B. Time-sharing OS.
- C. Real-time OS (RTOS).
- D. Network OS.

- **Answer: C | Explanation:** RTOS guarantees a response within a fixed time window.

17. What is a "System Call"?

- A. A call to customer support.
- B. A request to the hardware for data.
- C. An interface through which a program requests a service from the kernel.
- D. A method to reboot the computer.
- **Answer: C | Explanation:** System calls allow user programs to perform privileged actions safely.

18. "Dual-mode" operation is supported primarily by:

- A. The Operating System code.
- B. The Hardware/CPU.
- C. The User Application.
- D. The Antivirus software.
- **Answer: B | Explanation:** The CPU hardware must have bits to distinguish between User and Kernel modes.

19. Which of the following is NOT a core component of the OS?

- A. Processor Management.
- B. Memory Management.
- C. Database Management.
- D. File Management.
- **Answer: C | Explanation:** Database management is handled by a DBMS (application/server level software).

20. What does the "BIOS" (Basic Input/Output System) do?

- A. Manages user files.
- B. Initializes hardware and starts the booting process.
- C. Connects the computer to the internet.
- D. Compiles high-level code.
- **Answer: B | Explanation:** BIOS is firmware that runs the first instructions when power is applied.

21. A "Hosted" Operating System (like a VM) runs:

- A. Directly on the hardware.

- B. As a process within another "Host" OS.
- C. Inside the CPU cache.
- D. Only on servers.
- **Answer: B | Explanation:** Hosted hypervisors run on top of a standard OS like Windows or Linux.

22. Which architectural model uses a large, single-module kernel?

- A. Microkernel.
- B. Hybrid Kernel.
- C. Monolithic Kernel.
- D. Layered Kernel.
- **Answer: C | Explanation:** In monolithic kernels (like early Linux/Unix), all services are inside one big address space.

23. What is the "GUI" in OS?

- A. General Utility Interface.
- B. Graphical User Interface.
- C. Global User Index.
- D. Guarded Unit Interface.
- **Answer: B | Explanation:** GUI provides visual icons and windows for user interaction.

24. What is a "Resident" program?

- A. A program that lives on the hard drive.
- B. A program that remains in main memory at all times.
- C. A program that comes and goes as needed.
- D. A program that only runs at night.
- **Answer: B | Explanation:** The kernel is a resident program; it stays in RAM while the computer is on.

25. Which of the following is handled by the OS upon a "Power Failure"?

- A. Hardware restart.
- B. Orderly shutdown and state preservation (if UPS exists).
- C. Re-compiling broken code.
- D. None of the mentioned.
- **Answer: B | Explanation:** The OS must handle hardware interrupts like power failure signals gracefully.

26. Which type of OS is used in small devices like microwave ovens?

- A. Distributed OS.
- B. Embedded OS / RTOS.
- C. Network OS.
- D. Batch OS.
- **Answer: B | Explanation:** Embedded systems are specialized for single tasks with high reliability.

27. What is "Caching"?

- A. Deleting old files.
- B. Storing frequently accessed data in a faster memory layer.
- C. Encrypting passwords.
- D. Running two programs at once.
- **Answer: B | Explanation:** Caching improves performance by reducing access time to common data.

28. What is the "Interrupt Vector"?

- A. A list of all processes.
- B. A table of addresses for interrupt-handling routines.
- C. A type of network virus.
- D. A hardware signal to the CPU.
- **Answer: B | Explanation:** It maps the interrupt number to the specific code (ISR) that handles it.

29. Which OS is most common in supercomputers?

- A. Windows.
- B. Linux-based systems.
- C. MS-DOS.
- D. iOS.
- **Answer: B | Explanation:** Linux's performance and open-source nature make it ideal for high-performance clusters.

30. What is "Kernel Abstraction"?

- A. Hiding hardware complexity from the user.
- B. Deleting non-essential files.
- C. Increasing CPU speed.
- D. Compressing data.

- **Answer: A | Explanation:** Abstraction allows software to use resources (like a file) without knowing the physical details of the disk.
-

Part II: Process and Thread Management (Questions 31-70)

31. What is a "Process"?

- A. A program file on disk.
- B. A program in execution.
- C. A hardware device.
- D. A network signal.
- **Answer: B | Explanation:** A process is an active entity with a program counter, registers, and memory.

32. Which data structure contains the details of a process?

- A. Stack.
- B. Queue.
- C. Process Control Block (PCB).
- D. Heap.
- **Answer: C | Explanation:** The PCB stores PID, state, program counter, registers, etc.

33. What is a "Thread"?

- A. A heavy-weight process.
- B. A lightweight unit of execution within a process.
- C. A type of hard drive cable.
- D. A security protocol.
- **Answer: B | Explanation:** Threads share the process's code and data but have their own stack and registers.

34. In which state is a process when it is waiting for an I/O completion?

- A. Ready.
- B. Running.
- C. Waiting (or Blocked).
- D. New.
- **Answer: C | Explanation:** Blocked processes cannot run until the requested event finishes.

35. What is a "Context Switch"?

- A. Switching the computer off.
- B. Saving the state of a process and loading another.
- C. Changing the font of a document.
- D. Moving a window on the screen.
- **Answer: B | Explanation:** This is the overhead task that allows the OS to multitask.

36. A process that has finished execution but still has an entry in the PCB table is called:

- A. Orphan process.
- B. Zombie process.
- C. Daemon process.
- D. Thread.
- **Answer: B | Explanation:** A zombie process is waiting for its parent to read its exit status.

37. An "Orphan Process" is one whose:

- A. Parent has finished before it.
- B. Children have finished.
- C. Execution is stuck in a loop.
- D. Memory has been deallocated.
- **Answer: A | Explanation:** Orphans are usually adopted by the 'init' process in Linux.

38. Which of the following is NOT shared between threads of the same process?

- A. Code.
- B. Global Variables.
- C. Stack and Registers.
- D. Open Files.
- **Answer: C | Explanation:** Each thread needs its own stack for function calls and registers for its state.

39. "Inter-Process Communication" (IPC) allows processes to:

- A. Share hardware directly.
- B. Exchange data and synchronize their actions.
- C. Use the same PID.
- D. Delete each other.
- **Answer: B | Explanation:** IPC methods include Shared Memory, Pipes, and Message

Passing.

40. What is the main advantage of threads over processes?

- A. Better security.
- B. Faster creation and context switching.
- C. Full memory isolation.
- D. They don't require an OS to run.
- **Answer: B | Explanation:** Threads are lighter and share the same memory space, reducing overhead.

41. Which state comes after "Running" if a time slice expires?

- A. Waiting.
- B. Terminated.
- C. Ready.
- D. New.
- **Answer: C | Explanation:** In Round Robin, a process is put back in the Ready queue when its time is up.

42. The "Medium-term scheduler" is responsible for:

- A. Selecting processes from the disk to the RAM.
- B. Selecting processes from Ready queue to CPU.
- C. Swapping processes in and out of main memory.
- D. Managing the print queue.
- **Answer: C | Explanation:** It manages the degree of multiprogramming by swapping.

43. Which system call is used to wait for a child process to terminate?

- A. `fork()`
- B. `wait()`
- C. `signal()`
- D. `exec()`
- **Answer: B | Explanation:** `wait()` suspends the parent until the child process finishes.

44. What is a "System Daemon"?

- A. A virus.
- B. A process that runs in the background to handle system requests (like printing).
- C. A type of hardware controller.

- D. A user-level thread.
- **Answer: B | Explanation:** Daemons (like `httpd` or `syslogd`) provide continuous background services.

45. Which of the following can lead to a "Process Starvation"?

- A. Round Robin scheduling.
- B. Priority-based scheduling.
- C. FCFS.
- D. Multitasking.
- **Answer: B | Explanation:** High-priority tasks might keep a low-priority task from ever running.

46. What is the "Program Counter" inside a PCB?

- A. Counts the number of programs running.
- B. Indicates the address of the next instruction to be executed.
- C. Stores the value of an integer variable.
- D. Tracks the time used by the CPU.
- **Answer: B | Explanation:** It ensures the process can resume exactly where it was paused.

47. Which IPC method is the fastest for data sharing on the same machine?

- A. Message Queues.
- B. Sockets.
- C. Shared Memory.
- D. Pipes.
- **Answer: C | Explanation:** Since data is directly accessible in a shared RAM block, no kernel copying is needed.

48. What is the "Admission Scheduler"?

- A. Short-term scheduler.
- B. Long-term scheduler.
- C. Medium-term scheduler.
- D. None of the above.
- **Answer: B | Explanation:** It determines which jobs are admitted into the system for processing.

49. A "Process Control Block" (PCB) is created at:

- A. Process Termination.
- B. Process Creation.
- C. Context Switching.
- D. System Booting.
- **Answer: B | Explanation:** The OS creates a PCB as soon as a process is born.

50. What is a "System Call Interface"?

- A. The screen where you type commands.
- B. A table that links system calls to specific kernel routines.
- C. A network connection tool.
- D. A type of hardware bus.
- **Answer: B | Explanation:** It handles the redirection of user-level requests to the correct kernel code.

51. Which model maps many user-level threads to one kernel thread?

- A. One-to-One.
- B. Many-to-One.
- C. Many-to-Many.
- D. Two-level.
- **Answer: B | Explanation:** In this model, if one thread blocks, all threads in that process block.

52. "Kernel-level" threads are managed by:

- A. The User Library.
- B. The Operating System Kernel.
- C. The Compiler.
- D. The Application code.
- **Answer: B | Explanation:** The OS is aware of these threads and can schedule them across different cores.

53. In a multi-processor system, which thread model is most efficient for parallelism?

- A. Many-to-One.
- B. One-to-One.
- C. Single-threaded.
- D. Batch-to-One.
- **Answer: B | Explanation:** One-to-one allows multiple threads of the same process to

run on multiple CPUs simultaneously.

54. What is "Dispatch Latency"?

- A. Time for a process to reach the Ready queue.
- B. Time for the dispatcher to stop one process and start another.
- C. Total time a process spends in the system.
- D. Time to read a file from the disk.
- **Answer: B | Explanation:** It is a key metric for measuring OS responsiveness.

55. What happens when a process switches from "Running" to "Waiting"?

- A. It finished its time slice.
- B. It completed its execution.
- C. It made an I/O or event request.
- D. It was preempted by a higher priority task.
- **Answer: C | Explanation:** It enters a blocked state until the resource is ready.

56. A process is "Independent" if:

- A. It cannot affect or be affected by other processes.
- B. It has no parent.
- C. It runs only on one CPU.
- D. It uses no memory.
- **Answer: A | Explanation:** Independent processes do not share data or state with any other process.

57. What is "Degree of Multiprogramming"?

- A. The number of CPUs in the system.
- B. The number of processes currently in main memory.
- C. The speed of the processor.
- D. The size of the RAM.
- **Answer: B | Explanation:** It is managed by the long-term and medium-term schedulers.

58. Which of the following is an example of an "Interrupt-driven" I/O?

- A. CPU polling a device.
- B. A keyboard sending a signal when a key is pressed.
- C. A process running a loop.
- D. Deleting a file.

- **Answer: B | Explanation:** The device alerts the CPU only when it needs attention.

59. What is a "Signal" in Unix?

- A. A type of hardware wire.
- B. A software interrupt used to notify a process of an event.
- C. A way to increase CPU priority.
- D. A network packet.
- **Answer: B | Explanation:** Signals like `SIGKILL` or `SIGINT` communicate events to processes.

60. Which of these is a "Blocking" system call?

- A. `printf`
- B. `read()` from a slow disk/network.
- C. `variable = 10`
- D. `i++`
- **Answer: B | Explanation:** A blocking call stops the process execution until the operation completes.

Part III: CPU Scheduling (Questions 61-90)

61. Which scheduling algorithm uses "Time Quantum"?

- A. FCFS.
- B. Round Robin (RR).
- C. SJF.
- D. Priority.
- **Answer: B | Explanation:** RR is designed for time-sharing systems where every process gets a fixed slice.

62. Shortest Job First (SJF) scheduling is optimal in terms of:

- A. CPU Utilization.
- B. Average Waiting Time.
- C. Throughput.
- D. Fairness.
- **Answer: B | Explanation:** By finishing short jobs first, it minimizes the wait for the entire pool.

63. "Preemptive" scheduling means:

- A. A process cannot be stopped until it finishes.
- B. The OS can forcibly remove a process from the CPU.
- C. The process gives up the CPU voluntarily.
- D. Only one process can ever run.
- **Answer: B | Explanation:** This is essential for modern multitasking and real-time responsiveness.

64. What is the "Convoy Effect"?

- A. When many small processes run simultaneously.
- B. When many short processes wait for one long process in FCFS.
- C. A technique for faster booting.
- D. A security feature.
- **Answer: B | Explanation:** It occurs in FCFS when slow CPU-bound processes block fast I/O-bound ones.

65. Which algorithm is best suited for "Time-Sharing" systems?

- A. SJF.
- B. FCFS.
- C. Round Robin.
- D. LIFO.
- **Answer: C | Explanation:** It provides high responsiveness to multiple interactive users.

66. What is "Turnaround Time"?

- A. Time spent in the Ready queue.
- B. Total time from process submission to completion.
- C. Time from submission to first response.
- D. Time spent in I/O wait.
- **Answer: B | Explanation:** Formula: $\text{Turnaround Time} = \text{Finish Time} - \text{Arrival Time}$.

67. What is "Waiting Time"?

- A. Time spent running on CPU.
- B. Time spent in the Ready queue waiting for the CPU.
- C. Time spent waiting for I/O.
- D. Total time process exists.

- **Answer: B | Explanation:** Formula: $\text{Waiting Time} = \text{Turnaround Time} - \text{Burst Time}$.

68. The "Dispatcher" is responsible for:

- A. Choosing the process to run next.
- B. Giving control of the CPU to the process selected by the short-term scheduler.
- C. Swapping processes to disk.
- D. Handling system calls.
- **Answer: B | Explanation:** It performs the context switch and jumps to the correct program location.

69. What is "Response Time"?

- A. Total time to finish a job.
- B. Time from submission until the first response is produced.
- C. Time spent in I/O.
- D. Time to boot the OS.
- **Answer: B | Explanation:** It is critical for measuring the quality of interactive systems.

70. In "Multilevel Queue" scheduling, queues are often divided based on:

- A. Process size.
- B. Process type (e.g., Foreground vs. Background).
- C. Process owner.
- D. Process creation date.
- **Answer: B | Explanation:** High-priority interactive tasks usually go into foreground queues with RR.

71. What is the main problem with the "Shortest Remaining Time First" (SRTF) algorithm?

- A. It is very simple to implement.
- B. It can cause starvation of long processes.
- C. It has the highest waiting time.
- D. It doesn't support multitasking.
- **Answer: B | Explanation:** Continuous arrival of short jobs can permanently block long ones.

72. "Load Balancing" is a concept used in:

- A. Single-processor systems.

- B. Multiprocessor systems.
- C. Batch systems.
- D. Single-tasking systems.
- **Answer: B | Explanation:** It ensures tasks are distributed evenly across all available CPU cores.

73. What happens in "Priority Scheduling" if two processes have the same priority?

- A. The system crashes.
- B. They are handled using FCFS.
- C. They run at the same time on one CPU.
- D. The one with more memory runs first.
- **Answer: B | Explanation:** FCFS is the standard "tie-breaker" for most scheduling algorithms.

74. Which of the following is a non-preemptive algorithm?

- A. Round Robin.
- B. FCFS.
- C. SRTF.
- D. Multilevel Feedback Queue.
- **Answer: B | Explanation:** In FCFS, a process keeps the CPU until it finishes or voluntarily blocks.

75. What is a "Multilevel Feedback Queue"?

- A. A queue that never changes.
- B. A scheduling system where processes can move between queues based on their behavior.
- C. A queue for network packets only.
- D. A hardware component.
- **Answer: B | Explanation:** It penalizes CPU-bound jobs by moving them to lower-priority queues.

Part IV: Synchronization & Deadlocks (Questions 76-120)

76. A "Race Condition" occurs when:

- A. Two programmers code at the same time.
- B. Multiple threads access shared data and the final result depends on the order of execution.

- C. The CPU is running at maximum speed.
- D. A process is waiting for a resource.
- **Answer: B | Explanation:** This is a core problem in concurrent programming.

77. What is "Mutual Exclusion"?

- A. Multiple processes using the same resource simultaneously.
- B. Ensuring only one process uses a shared resource at a time.
- C. Excluding high-priority tasks from the CPU.
- D. Deleting all files in a folder.
- **Answer: B | Explanation:** It is the first of the 4 conditions for deadlock.

78. A "Critical Section" is:

- A. A segment of the kernel that cannot be changed.
- B. Code where a process accesses shared resources.
- C. The most important part of the hard drive.
- D. A segment of code that runs during booting.
- **Answer: B | Explanation:** Critical sections must be protected to prevent data corruption.

79. What is a "Deadlock"?

- A. A process that has finished execution.
- B. A state where processes are blocked forever, each waiting for a resource held by another.
- C. A hardware failure in the CPU.
- D. A network connection timeout.
- **Answer: B | Explanation:** Deadlocks lead to system hangs where no progress is possible.

80. Which of these is NOT a condition for Deadlock?

- A. Mutual Exclusion.
- B. Circular Wait.
- C. Preemption.
- D. Hold and Wait.
- **Answer: C | Explanation:** Deadlock requires "No Preemption." If preemption was allowed, deadlocks could be broken.

81. What does "Hold and Wait" mean?

- A. A process holds all resources and finishes execution.
- B. A process holds at least one resource and is waiting for more.
- C. A process waits for the CPU to start.
- D. A process is deleted from the queue.
- **Answer: B | Explanation:** This condition allows the "Circular Wait" chain to form.

82. The "Banker's Algorithm" is used for Deadlock:

- A. Prevention.
- B. Avoidance.
- C. Detection.
- D. Recovery.
- **Answer: B | Explanation:** It checks for a "Safe State" before granting resource requests.

83. A "Safe State" is a state where:

- A. No viruses are present.
- B. The system can allocate resources to all processes without a deadlock.
- C. The computer is turned off.
- D. All processes have finished.
- **Answer: B | Explanation:** Safe states guarantee that a deadlock can always be avoided.

84. "Circular Wait" can be prevented by:

- A. Increasing the number of resources.
- B. Imposing a linear ordering of resources.
- C. Using more threads.
- D. Restarting the system.
- **Answer: B | Explanation:** If processes must request resources in order (1 then 2), a cycle cannot form.

85. What is a "Semaphore"?

- A. A hardware flag.
- B. An integer variable used for synchronization via atomic `wait()` and `signal()` calls.
- C. A type of process scheduler.
- D. A file compression tool.
- **Answer: B | Explanation:** Semaphores are fundamental tools for managing access to

critical sections.

86. A "Binary Semaphore" is also known as a:

- A. Monitor.
- B. Mutex.
- C. Pipe.
- D. Stack.
- **Answer: B | Explanation:** Mutexes (Mutual Exclusion) only allow values 0 or 1.

87. What is "Priority Inversion"?

- A. Increasing priority through aging.
- B. A high-priority process being blocked by a lower-priority process holding a resource.
- C. Changing the user's priority.
- D. When a process finishes too fast.
- **Answer: B | Explanation:** It can lead to critical failures in real-time systems if not managed.

88. "Resource Allocation Graph" (RAG) is used to:

- A. Allocate RAM to processes.
- B. Detect deadlocks (cycles in the graph).
- C. Measure CPU speed.
- D. Design the file system.
- **Answer: B | Explanation:** If a RAG has a cycle and single-instance resources, a deadlock exists.

89. What is "Bounded Waiting"?

- A. A process waiting for exactly 10 seconds.
- B. A guarantee that a process will eventually enter its critical section after a finite wait.
- C. Waiting for a resource that doesn't exist.
- D. A process that never waits.
- **Answer: B | Explanation:** It ensures fairness and prevents starvation in critical section solutions.

90. A "Spinlock" is a type of lock that:

- A. Puts the process to sleep.
- B. Uses "Busy Waiting" (looping) until the lock is free.

- C. Automatically deletes the process.
 - D. Is only used in mobile OS.
 - **Answer: B | Explanation:** Spinlocks are fast for very short wait times but waste CPU cycles if the wait is long.
-

Part V: Memory Management (Questions 121-160)

91. What is the purpose of "Paging"?

- A. To increase disk speed.
- B. To allow non-contiguous allocation of physical memory.
- C. To organize files into folders.
- D. To protect the kernel.
- **Answer: B | Explanation:** Paging splits process memory into blocks, allowing them to be scattered in RAM.

92. "Logical Address" is generated by:

- A. The RAM.
- B. The CPU during program execution.
- C. The Hard drive.
- D. The BIOS.
- **Answer: B | Explanation:** Programs use these virtual addresses, which the MMU translates to physical ones.

93. What is "External Fragmentation"?

- A. Unused space inside an allocated block.
- B. Small, non-contiguous holes in memory that cannot satisfy a request.
- C. Space wasted on the hard drive.
- D. A virus from an external USB.
- **Answer: B | Explanation:** It is a major drawback of contiguous memory allocation.

94. "Internal Fragmentation" occurs when:

- A. Total memory is less than required.
- B. Allocated memory is slightly larger than requested memory (e.g., in paging).
- C. The system is turned off.
- D. Files are deleted from disk.
- **Answer: B | Explanation:** It happens because memory is often allocated in fixed-size

blocks (pages/frames).

95. The "Memory Management Unit" (MMU) is:

- A. Software in the OS.
- B. A hardware component that translates addresses.
- C. A type of RAM.
- D. Part of the file system.
- **Answer: B | Explanation:** It provides the high-speed translation needed for paging and protection.

96. What is "Virtual Memory"?

- A. A simulation of a computer.
- B. A technique that allows programs to run using disk space as extra RAM.
- C. Cloud-based storage.
- D. Memory used by virtual reality headsets.
- **Answer: B | Explanation:** It allows the execution of processes larger than physical RAM.

97. A "Page Fault" happens when:

- A. The CPU has an error.
- B. A requested page is not in physical RAM.
- C. A file is not found on disk.
- D. The computer restarts.
- **Answer: B | Explanation:** The OS must then load the missing page from swap space on the disk.

98. What is "Thrashing"?

- A. A process running very fast.
- B. A state where the system spends more time swapping pages than executing.
- C. Deleting files too quickly.
- D. A hardware failure.
- **Answer: B | Explanation:** It occurs when the system's "working set" of pages exceeds its physical memory.

99. Which page replacement algorithm replaces the page that hasn't been used for the longest time?

- A. FIFO.
- B. Optimal.
- C. LRU (Least Recently Used).
- D. LIFO.
- **Answer: C | Explanation:** It is based on the principle of locality of reference.

100. "Segmentation" is a memory management scheme that supports:

- A. Fixed-size blocks.
- B. The programmer's view of memory (logical units like stacks, code, data).
- C. Higher disk speeds.
- D. Network security.
- **Answer: B | Explanation:** Unlike paging, segmentation uses variable-sized logical blocks.

Part VI: File Systems & I/O (Questions 161-200)

101. An "Inode" in Unix contains:

- A. The filename.
- B. File metadata (size, owner, permissions, block pointers).
- C. The physical location of the CPU.
- D. A list of all websites visited.
- **Answer: B | Explanation:** Inodes represent the file itself; filenames are stored in directory files.

102. What is "Seek Time"?

- A. Time to read data from a disk sector.
- B. Time for the disk head to move to the correct cylinder/track.
- C. Time to find a file in a folder.
- D. Total processing time.
- **Answer: B | Explanation:** Seek time is the most significant delay in mechanical hard drives.

103. "Direct Memory Access" (DMA) is used to:

- A. Let the CPU access RAM faster.
- B. Allow high-speed I/O devices to transfer data to RAM without using the CPU for every byte.
- C. Protect memory from viruses.

- D. Increase disk space.
- **Answer: B | Explanation:** DMA significantly reduces CPU overhead during large data transfers.

104. What is "Disk Scheduling"?

- A. Planning when to buy a new disk.
- B. Managing the order in which I/O requests are serviced to minimize seek time.
- C. Formatting a hard drive.
- D. Running a backup.
- **Answer: B | Explanation:** Algorithms like SCAN and SSTF help optimize mechanical disk movement.

105. "Belady's Anomaly" is associated with which algorithm?

- A. LRU.
- B. FIFO.
- C. Optimal.
- D. SJF.
- **Answer: B | Explanation:** It describes a situation where more page frames lead to more page faults in FIFO.

106. What is the "Dirty Bit" used for in Paging?

- A. To identify corrupt pages.
- B. To indicate if a page has been modified (so it must be written to disk before being replaced).
- C. To track user login time.
- D. To increase CPU speed.
- **Answer: B | Explanation:** It avoids unnecessary disk writes for pages that haven't changed.

107. Which RAID level uses "Mirroring" for redundancy?

- A. RAID 0.
- B. RAID 1.
- C. RAID 5.
- D. RAID 10.
- **Answer: B | Explanation:** RAID 1 writes the same data to two disks simultaneously.

108. What is a "Hard Link"?

- A. A physical cable.
- B. A directory entry that points to the same inode as another file.
- C. A shortcut that breaks if the original file is moved.
- D. A file that cannot be deleted.
- **Answer: B | Explanation:** Hard links are indistinguishable from the original file at the inode level.

109. What is "Rotational Latency"?

- A. Time for the CPU to cycle.
- B. Time for the disk to rotate the desired sector under the head.
- C. Time to restart the OS.
- D. The speed of the mouse.
- **Answer: B | Explanation:** It is a component of the total disk access time.

110. "Buffering" is primarily used to:

- A. Match speed differences between two devices.
- B. Increase the size of the hard drive.
- C. Encrypt data.
- D. Hide files from the user.
- **Answer: A | Explanation:** Buffers temporarily hold data to handle variations in data transfer rates.

111. What is the "Root Directory"?

- A. The top-level directory in a file system hierarchy.
- B. A folder for system updates.
- C. The main directory of a website.
- D. A list of all hardware drivers.
- **Answer: A | Explanation:** In Unix, it is represented by `/`.

112. Which of the following is a characteristic of a "Network OS"?

- A. It runs on a single computer without network cards.
- B. It provides features to manage users and share files across a network.
- C. It does not have a kernel.
- D. It is only used for gaming.
- **Answer: B | Explanation:** NOS (like Windows Server) is built for connectivity and

sharing.

113. What is a "Swap Space"?

- A. RAM space.
- B. Space on the hard drive used to hold process data when RAM is full.
- C. Space for system logs.
- D. A cloud storage area.
- **Answer: B | Explanation:** It is the backing store for virtual memory operations.

114. "Fragmentation" in a file system occurs when:

- A. Files are deleted.
- B. A file is stored in non-contiguous blocks on the disk.
- C. The CPU is overloaded.
- D. The power fails.
- **Answer: B | Explanation:** Fragmentation slows down read speeds in traditional hard drives.

115. What is "Direct Access" (or Random Access)?

- A. Accessing data in a fixed sequence.
- B. Accessing data in any order immediately.
- C. Accessing data only after a password.
- D. Accessing data from the cloud.
- **Answer: B | Explanation:** Modern hard drives and SSDs support direct access, unlike magnetic tapes.

116. Which of the following is NOT a valid file extension?

- A. .txt
- B. .exe
- C. .mem (for system memory)
- D. .kernel
- **Answer: D | Explanation:** While kernels exist as files, .kernel is not a standard functional extension.

117. What is "Mounting" a file system?

- A. Putting the hard drive inside the case.
- B. Making a file system available for access at a specific location in the directory tree.

- C. Deleting a partition.
- D. Increasing the RAM.
- **Answer: B | Explanation:** The `mount` command attaches a disk or partition to the main OS file tree.

118. What is "Journaling" in a file system?

- A. Writing a diary of user activities.
- B. Keeping a log of changes before applying them to avoid corruption during crashes.
- C. Organizing files by date.
- D. Compressing old files.
- **Answer: B | Explanation:** It improves the reliability and recovery speed of modern file systems like NTFS or ext4.

119. Which disk scheduling algorithm is often called the "Elevator" algorithm?

- A. FCFS.
- B. SSTF.
- C. SCAN.
- D. LIFO.
- **Answer: C | Explanation:** Like an elevator, it services all requests in one direction before reversing.

120. In an "Indexed Allocation" of files:

- A. Files are stored contiguously.
- B. Each file has its own index block containing pointers to its data blocks.
- C. Files are stored as a linked list.
- D. There are no files, only raw data.
- **Answer: B | Explanation:** It supports direct access and eliminates external fragmentation.